

# ELI — Inference And Hardware

ELI v2.3.1

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## ELI Inference & Hardware Boot

How ELI loads a model, talks to it, and adapts to whatever machine it's on. The inference path is model-agnostic (see memory eli-model-agnostic); the boot path is hardware-adaptive. Files in eli/cognition/ and eli/core/.

### Inference (cognition/gguf\_inference.py, 2.7k LOC + inference\_broker.py)

- **Model resolution (get\_model\_path):** ELI\_GGUF\_MODEL\_PATH env → model\_path/custom\_model\_path/bundled\_model\_path settings keys. No baked model; empty default.
- **load\_model(force\_reload):** resolves n\_ctx (env → settings → config.get\_gguf\_n\_ctx()), caps it to vision\_coresident\_text\_ctx when a co-resident vision model is loaded; resolves n\_gpu\_layers similarly.
- **Graceful GPU-layer fallback:** if context allocation fails at the requested layer count, it retries with fewer layers / without flash-attention rather than crashing — keeps the model + n\_ctx and degrades GPU offload instead.
- **Chat templating is family-aware:** \_is\_mistral\_model / \_is\_chatml\_model / \_is\_llama\_model sniff the filename to pick the right prompt format. This is *adaptation* to whatever model you load (like the ctx table), not a hardcoded model — but it is filename-based, so an unrecognised naming scheme falls back to a default template.
- **Serialization:** all calls hold \_LLM\_CALL\_LOCK (a native RLock from runtime/native\_locks) — llama\_cpp is not safe under concurrent calls; this is also what vision hot-swap and the ambient daemon coordinate on.
- **Live control:** get\_live\_runtime\_override, unload\_model, reload\_model let the GUI swap models / change settings without a full restart.
- **InferenceBroker** (inference\_broker.py): the higher-level infer() / gguf\_ready abstraction the orchestrator, engine, and ReAct loop call, so callers don't touch gguf\_inference directly.

## Hardware profiling (core/hardware\_profile.py, 1232 LOC)

Free-VRAM-aware sizing: - HardwareProfile dataclass tracks **free** vs total VRAM (free is what matters for whether a profile actually loads). - `_kv_cache_mb(n_ctx, n_layers)` — KV-cache cost. - `_compute_graph_reserve_mb(n_ctx, batch)` — the **model-agnostic** compute buffer estimate (256MB + 24MB/1K ctx + 1.5MB/batch), reserved so a profile that loads cleanly doesn't then hard-crash on the first decode when the lazy compute buffer pushes VRAM over the limit. (This was a real crash class.) - `_layers_for_size, ModelRecommendation` — pick offload layers from model size and free VRAM.

## Boot optimizer (core/startupHardware\_optimizer.py, 655 LOC)

Runs at startup, writes artifacts/runtimeHardware\_profile.json: - `detect_ram_gb, detect_cpu_name, detect_nvidia_gpus` (+ `detect_other_gpus` fallback), `select_gpu`. - `find_model(settings)` — locate the GGUF. - `train_ctx_for_model(model_path)` — the filename→context table (deepseek/llama-3.1/phi/gemma-2 → 128K; qwen2.5/mistral-7b → 32K; older → 8K; unknown → 32768). This is the core of model-agnostic context sizing. - `estimate_layers, layer_mb` — VRAM-fit layer count.

## Settings (core/runtime\_settings.py, 1134 LOC)

- DEFAULTS (the full settings schema) + `ENV_TO_KEY` (env-var overrides).
- `load_settings / save_settings / update_settings`.
- **Redistribution-aware:** `_migrate_legacy_keys` (schema evolution), `_resolve_relative_model_paths` + `_heal_model_paths` (fix stale absolute paths when the project moves machines), and `_portable_settings_for_storage` (strip machine-specific values before storage). The intent to keep settings portable across machines is built in — though personal values like `user_name` can still end up tracked (see the settings.json commit note).

## Paths (core/paths.py, 601 LOC)

Dev-vs-packaged path resolution: `is_frozen / is_dev_mode, _find_project_root`, then `data_dir/config_dir/cache_dir/memory_db_dir/artifacts_dir/user_db_path/agent_db_path/memory_db_path`. Source checkouts use project-local artifacts/+config/; packaged installs use platformdirs. One import surface (`get_paths`) so nothing hardcodes locations.

## Honest assessment

- **Strong:** genuinely adaptive and agnostic — free-VRAM-aware sizing, the compute-buffer reservation that prevents first-decode crashes, graceful GPU-layer fallback, filename→ctx adaptation, env/settings/relative-path healing for moving between machines, and a single broker + lock so concurrency is correct. This is mature, hard-won infrastructure.
- **Weak / watch:**
  1. **Filename-based family detection** (chat template + ctx) is fragile: a model with an unconventional filename gets a default template + 32768 ctx, which can be wrong (mis-templated output, or ctx overflow on a small model). A metadata/GGUF-header probe would be more robust than string matching.
  2. **Settings sprawl** — DEFAULTS is large with several overlapping keys (`n_gpu_layers/gpu_layers, n_ctx/context_size`) and migration logic, echoing the schema churn seen in memory/.
  3. VRAM heuristics are empirically tuned around an 8GB card; very different hardware (24GB+, or CPU-only) leans on the conservative fallbacks rather than tuned values.
  4. `_portable_settings_for_storage` exists but isn't fully preventing personal values (e.g. `user_name`) from being persisted/committed — worth tightening for redistribution.

## Update — 2026-06-09 (STT no longer starves the main model)

- faster-whisper preloaded on CUDA **before** the main GGUF autotuned, claiming ~2 GB so on an 8 GB card the main model only got `gpu_layers=11` (`free_vram=4083 MB`) → 23-59 s turns. STT is now VRAM-aware (`local_whisper_stt`: GPU only on  $\geq 12$  GB cards, else CPU), so the main model reclaims the GPU — `gpu_layers=99` / `ctx=20480` with `free_vram` ~7 GB, turns back to 1-12 s. `ELI_WHISPER_DEVICE` overrides; `ELI_WHISPER_GPU_MIN_MB` tunes the threshold. See `perception.md`.